

DSA MCQs

1. Which step comes first in problem solving?

- A) Implementation
- B) Problem identification
- C) Testing
- D) Optimization

Answer: B

2. What is the main goal of computational thinking?

- A) Write code quickly
- B) Solve problems systematically
- C) Use only loops
- D) Avoid algorithms

Answer: B

3. Which data structure follows LIFO principle?

- A) Queue
- B) Stack
- C) Array
- D) Tree

Answer: B

4. What is time complexity used to measure?

- A) Memory usage
- B) Execution time growth with input size
- C) CPU temperature
- D) Output correctness

Answer: B

5. What is the output of this stack operation?

Push(10), Push(20), Pop()

- A) 10
- B) 20

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- C) 0
- D) Error

Answer: B

6. Which data structure uses FIFO principle?

- A) Stack
- B) Queue
- C) Tree
- D) Graph

Answer: B

7. What is circular queue advantage?

- A) Wastes memory
- B) Reuses freed space
- C) No overflow possible
- D) No dequeue operation

Answer: B

8. Which structure is best for recursion?

- A) Queue
- B) Stack
- C) Graph
- D) Heap

Answer: B

9. What is base condition in recursion?

- A) Infinite loop condition
- B) Condition to stop recursion
- C) Sorting condition
- D) Error condition

Answer: B

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10. What is worst-case complexity of linear search?

- A) $O(1)$
- B) $O(\log n)$
- C) $O(n)$
- D) $O(n^2)$

Answer: C

11. Which traversal visits root first?

- A) Inorder
- B) Postorder
- C) Preorder
- D) Level order

Answer: C

12. Which traversal uses queue?

- A) DFS
- B) BFS
- C) Inorder
- D) Postorder

Answer: B

13. Which tree ensures balanced height?

- A) Binary tree
- B) AVL tree
- C) Skewed tree
- D) Linked list

Answer: B

14. What is BST property?

- A) $\text{Left} > \text{Root} > \text{Right}$
- B) $\text{Left} < \text{Root} < \text{Right}$
- C) Random order
- D) Equal values only

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Answer: B

15. Which sorting algorithm is divide and conquer?

- A) Bubble sort
- B) Insertion sort
- C) Quick sort
- D) Selection sort

Answer: C

16. Worst-case of Quick sort is:

- A) $O(n \log n)$
- B) $O(\log n)$
- C) $O(n^2)$
- D) $O(n)$

Answer: C

17. Which sorting is stable by default?

- A) Quick sort
- B) Merge sort
- C) Selection sort
- D) Heap sort

Answer: B

18. Binary search requires:

- A) Unsorted array
- B) Sorted array
- C) Linked list
- D) Tree only

Answer: B

19. Complexity of binary search:

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- A) $O(n)$
- B) $O(\log n)$
- C) $O(n^2)$
- D) $O(1)$

Answer: B

20. Hashing is used for:

- A) Sorting
- B) Fast searching
- C) Recursion
- D) Tree traversal

Answer: B

21. Collision in hashing means:

- A) Same key, same value
- B) Two keys map to same index
- C) Empty table
- D) Overflow only

Answer: B

22. Linear probing resolves collision by:

- A) Random placement
- B) Next available slot
- C) Deleting element
- D) Rehashing always

Answer: B

23. Which graph traversal uses stack?

- A) BFS
- B) DFS
- C) Dijkstra
- D) Floyd-Warshall

Answer: B

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24. BFS uses:

- A) Stack
- B) Queue
- C) Heap
- D) Tree

Answer: B

25. Adjacency matrix is:

- A) Sparse friendly
- B) Dense graph representation
- C) Linked list structure
- D) Tree structure

Answer: B

26. Space complexity of adjacency matrix:

- A) $O(n)$
- B) $O(n^2)$
- C) $O(\log n)$
- D) $O(1)$

Answer: B

27. Dijkstra algorithm finds:

- A) Longest path
- B) Shortest path
- C) Tree height
- D) Sorting order

Answer: B

28. Kruskal algorithm is used for:

- A) BFS
- B) MST

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- C) Sorting
- D) Searching

Answer: B

29. Prim's algorithm is for:

- A) Shortest path
- B) MST
- C) Sorting
- D) Recursion

Answer: B

30. Greedy algorithm makes:

- A) Global optimal choice always
- B) Local optimal choice
- C) Random choice
- D) Backtracking choice

Answer: B

31. Dynamic programming uses:

- A) Recursion without memory
- B) Overlapping subproblems
- C) Random selection
- D) Graph only

Answer: B

32. Backtracking is used for:

- A) Sorting
- B) Exploring all possibilities
- C) Hashing
- D) Searching only

Answer: B

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33. Best case complexity means:

- A) Maximum time
- B) Minimum time
- C) Average time
- D) Random time

Answer: B

34. Space complexity refers to:

- A) CPU usage
- B) Memory usage
- C) Disk speed
- D) Output size

Answer: B

35. Which data structure is non-linear?

- A) Array
- B) Stack
- C) Tree
- D) Queue

Answer: C

36. Linked list advantage over array:

- A) Fixed size
- B) Dynamic size
- C) Faster indexing
- D) Less memory always

Answer: B

37. Circular linked list last node points to:

- A) NULL
- B) First node
- C) Middle node
- D) Random node

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Answer: B

38. DFS can be implemented using:

- A) Queue
- B) Stack
- C) Array only
- D) Heap

Answer: B

39. Which algorithm is brute force?

- A) Quick sort
- B) Selection sort
- C) Merge sort
- D) Dijkstra

Answer: B

40. Complexity of merge sort:

- A) $O(n^2)$
- B) $O(n \log n)$
- C) $O(n)$
- D) $O(\log n)$

Answer: B

41. Which structure supports priority processing?

- A) Stack
- B) Queue
- C) Heap
- D) Array

Answer: C

42. Heap is used in:

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- A) BFS
- B) Priority queue
- C) Linked list
- D) Stack only

Answer: B

43. Which is NOT a graph type?

- A) Directed
- B) Undirected
- C) Weighted
- D) Linear

Answer: D

44. Floyd-Warshall is used for:

- A) Single source shortest path
- B) All pairs shortest path
- C) Sorting
- D) Tree traversal

Answer: B

45. In BFS, nodes are visited:

- A) Depth-wise
- B) Level-wise
- C) Randomly
- D) Reverse order

Answer: B

46. Which is faster in average case?

- A) Bubble sort
- B) Quick sort
- C) Selection sort
- D) Insertion sort

Answer: B

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47. Recursive function always needs:

- A) Loop
- B) Base condition
- C) Array
- D) Pointer

Answer: B

48. Stack overflow occurs due to:

- A) Queue overflow
- B) Infinite recursion
- C) Sorting
- D) Hashing

Answer: B

49. Which is a linear data structure?

- A) Tree
- B) Graph
- C) Array
- D) BST

Answer: C

50. Which algorithm is best for sparse graph MST?

- A) Prim
- B) Kruskal
- C) BFS
- D) DFS

Answer: B

51. What is the time complexity of BFS on a graph represented using adjacency list?

- A) $O(V^2)$
- B) $O(V + E)$

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- C) $O(E^2)$
- D) $O(\log V)$

Answer: B

52. In DFS, recursion stack primarily simulates:

- A) Queue
- B) Heap
- C) Stack
- D) Array sorting

Answer: C

53. Which graph traversal is better for finding shortest path in unweighted graph?

- A) DFS
- B) BFS
- C) Dijkstra
- D) Prim

Answer: B

54. What is the key reason Quick Sort performs poorly in worst case?

- A) Too many swaps
- B) Unbalanced partitioning
- C) Extra memory usage
- D) Lack of recursion

Answer: B

55. Which sorting algorithm uses divide and conquer AND extra memory?

- A) Quick sort
- B) Merge sort
- C) Selection sort
- D) Bubble sort

Answer: B

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56. Worst-case of merge sort is:

- A) $O(n^2)$
- B) $O(n \log n)$
- C) $O(\log n)$
- D) $O(n)$

Answer: B

57. Which sorting algorithm is adaptive (performs better on nearly sorted data)?

- A) Bubble sort
- B) Merge sort
- C) Heap sort
- D) Quick sort

Answer: A

58. Which recurrence relation represents merge sort?

- A) $T(n) = T(n-1) + n$
- B) $T(n) = 2T(n/2) + n$
- C) $T(n) = T(n/2) + 1$
- D) $T(n) = n^2$

Answer: B

59. Which algorithm technique is used in binary search?

- A) Greedy
- B) Divide and conquer
- C) Backtracking
- D) Dynamic programming

Answer: B

60. Time complexity of binary search recursion:

- A) $O(n)$
- B) $O(\log n)$
- C) $O(n \log n)$
- D) $O(1)$

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Answer: B

61. Which case makes Quick Sort worst?

- A) Random pivot
- B) Sorted input with first element pivot
- C) Balanced partition
- D) Reverse sorted with random pivot

Answer: B

62. Which data structure is used in recursion internally?

- A) Queue
- B) Stack
- C) Tree
- D) Graph

Answer: B

63. Memoization is used in:

- A) Greedy
- B) Dynamic programming
- C) BFS
- D) Sorting

Answer: B

64. DP differs from recursion by:

- A) No recursion
- B) Stores intermediate results
- C) Uses graph
- D) Uses sorting

Answer: B

65. Which is top-down DP approach?

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- A) Iteration
- B) Memoization
- C) Greedy
- D) BFS

Answer: B

66. Which is bottom-up DP?

- A) Recursion
- B) Tabulation
- C) Backtracking
- D) DFS

Answer: B

67. Which problem is classic DP example?

- A) BFS traversal
- B) Fibonacci series
- C) Binary search
- D) Heap sort

Answer: B

68. Time complexity of naive Fibonacci recursion:

- A) $O(n)$
- B) $O(\log n)$
- C) $O(2^n)$
- D) $O(n \log n)$

Answer: C

69. Optimized Fibonacci using DP gives:

- A) $O(n^2)$
- B) $O(n)$
- C) $O(\log n)$
- D) $O(2^n)$

Answer: B

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70. Which graph algorithm uses greedy approach?

- A) DFS
- B) Dijkstra
- C) Floyd Warshall
- D) BFS

Answer: B

71. Floyd-Warshall complexity is:

- A) $O(VE)$
- B) $O(V^3)$
- C) $O(E^2)$
- D) $O(V^2 \log V)$

Answer: B

72. Which algorithm is used for all-pairs shortest path?

- A) Dijkstra
- B) Bellman-Ford
- C) Floyd-Warshall
- D) Kruskal

Answer: C

73. Bellman-Ford can handle:

- A) Negative weights
- B) Only positive weights
- C) No weights
- D) Trees only

Answer: A

74. Cycle detection in directed graph uses:

- A) BFS only
- B) DFS recursion stack

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- C) Sorting
- D) DP

Answer: B

75. Topological sorting applies to:

- A) Undirected graphs
- B) DAG
- C) Cyclic graphs
- D) Trees only

Answer: B

76. Kruskal algorithm uses:

- A) Queue
- B) Stack
- C) Disjoint set
- D) Heap only

Answer: C

77. Prim's algorithm is:

- A) Edge-based
- B) Vertex-based
- C) Random
- D) DFS-based

Answer: B

78. Which sorting algorithm is NOT comparison-based?

- A) Quick sort
- B) Merge sort
- C) Radix sort
- D) Heap sort

Answer: C

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79. Time complexity of heap sort:

- A) $O(n^2)$
- B) $O(n \log n)$
- C) $O(n)$
- D) $O(\log n)$

Answer: B

80. Heap construction complexity:

- A) $O(n \log n)$
- B) $O(n)$
- C) $O(\log n)$
- D) $O(n^2)$

Answer: B

81. Which traversal gives sorted output in BST?

- A) Preorder
- B) Inorder
- C) Postorder
- D) Level order

Answer: B

82. Height of balanced BST is:

- A) $O(n)$
- B) $O(\log n)$
- C) $O(n^2)$
- D) $O(1)$

Answer: B

83. Worst-case BST search becomes:

- A) $O(\log n)$
- B) $O(n)$
- C) $O(1)$
- D) $O(n \log n)$

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Answer: B

84. AVL tree maintains:

- A) Sorted array
- B) Balance factor
- C) Hashing
- D) Queue order

Answer: B

85. Rotation in AVL tree is used for:

- A) Searching
- B) Balancing
- C) Sorting
- D) Deleting

Answer: B

86. Which graph is best represented using adjacency list?

- A) Dense graph
- B) Sparse graph
- C) Complete graph only
- D) Tree only

Answer: B

87. Space complexity of adjacency list:

- A) $O(V^2)$
- B) $O(V + E)$
- C) $O(E^2)$
- D) $O(\log V)$

Answer: B

88. DFS traversal uses:

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- A) FIFO
- B) LIFO
- C) Random access
- D) Sorting

Answer: B

89. BFS traversal uses:

- A) Stack
- B) Queue
- C) Heap
- D) Tree

Answer: B

90. Which is better for shortest path in weighted graph?

- A) BFS
- B) DFS
- C) Dijkstra
- D) Selection sort

Answer: C

91. Which algorithm is greedy?

- A) Merge sort
- B) Dijkstra
- C) DFS
- D) Backtracking

Answer: B

92. Backtracking is similar to:

- A) DFS with pruning
- B) BFS
- C) Sorting
- D) Hashing

Answer: A

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93. Which is NOT a DP property?

- A) Overlapping subproblems
- B) Optimal substructure
- C) Greedy choice
- D) Memoization

Answer: C

94. Which algorithm is brute force?

- A) Quick sort
- B) Bubble sort
- C) Binary search
- D) Dijkstra

Answer: B

95. Which is most efficient for large sorted data search?

- A) Linear search
- B) Binary search
- C) DFS
- D) Bubble sort

Answer: B

96. Worst-case space complexity of recursion:

- A) $O(1)$
- B) $O(n)$
- C) $O(\log n)$
- D) $O(n^2)$

Answer: B

97. Tail recursion optimization reduces:

- A) Time only
- B) Stack usage

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- C) Memory only
- D) Sorting time

Answer: B

98. Which technique avoids recomputation?

- A) Greedy
- B) Memoization
- C) DFS
- D) BFS

Answer: B

99. Which is true for greedy algorithm?

- A) Always optimal
- B) Locally optimal choice
- C) Uses recursion only
- D) Uses sorting only

Answer: B

100. Which problem is best solved by backtracking?

- A) Sorting
- B) N-Queens
- C) BFS
- D) Binary search

Answer: B

101. Which graph algorithm detects shortest path with negative edges?

- A) Dijkstra
- B) Bellman-Ford
- C) Prim
- D) Kruskal

Answer: B

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102. Which structure is used in BFS?

- A) Stack
- B) Queue
- C) Tree
- D) Graph

Answer: B

103. DFS time complexity:

- A) $O(V + E)$
- B) $O(V^2)$
- C) $O(E^2)$
- D) $O(\log V)$

Answer: A

104. Which sorting is best for small datasets?

- A) Merge sort
- B) Insertion sort
- C) Heap sort
- D) Quick sort

Answer: B

105. Which algorithm is not divide and conquer?

- A) Merge sort
- B) Quick sort
- C) Binary search
- D) Bubble sort

Answer: D

106. Which data structure is best for BFS implementation?

- A) Stack
- B) Queue
- C) Heap
- D) Array

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Answer: B

107. Which is correct for recursion?

- A) Must have loop
- B) Must have base case
- C) Must use array
- D) Must use graph

Answer: B

108. Which algorithm uses pivot?

- A) Merge sort
- B) Quick sort
- C) Heap sort
- D) BFS

Answer: B

109. Which is most space efficient sorting?

- A) Merge sort
- B) Quick sort
- C) Bubble sort
- D) Selection sort

Answer: B

110. Which algorithm is best for dense graphs MST?

- A) Kruskal
- B) Prim
- C) BFS
- D) DFS

Answer: B

111. Which technique reduces exponential recursion?

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- A) Greedy
- B) DP
- C) BFS
- D) Sorting

Answer: B

112. Which traversal is recursive by nature?

- A) BFS
- B) DFS
- C) Level order
- D) None

Answer: B

113. Which algorithm is used for minimum spanning tree?

- A) DFS
- B) BFS
- C) Prim
- D) Binary search

Answer: C

114. Which sorting algorithm is stable and recursive?

- A) Heap sort
- B) Merge sort
- C) Quick sort
- D) Selection sort

Answer: B

115. Which is true about binary tree?

- A) Each node has 3 children
- B) At most 2 children
- C) No children allowed
- D) Only root exists

Answer: B

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116. Which structure is used in recursion call stack?

- A) Queue
- B) Stack
- C) Heap
- D) Graph

Answer: B

117. Which algorithm is best for topological sorting?

- A) DFS
- B) BFS only
- C) Dijkstra
- D) Prim

Answer: A

118. Which is used in dynamic programming?

- A) Storing results
- B) Random choice
- C) Sorting only
- D) Graph only

Answer: A

119. Which is NOT a graph traversal?

- A) BFS
- B) DFS
- C) Dijkstra
- D) Inorder

Answer: D

120. Which technique solves optimization problems efficiently?

- A) Brute force
- B) DP

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- C) Random search
- D) Linear scan

Answer: B

121. Which data structure is most suitable for implementing recursion explicitly?

- A) Queue
- B) Stack
- C) Heap
- D) Graph

Answer: B

122. What is the space complexity of recursive Fibonacci without memoization?

- A) $O(1)$
- B) $O(n)$
- C) $O(2^n)$
- D) $O(\log n)$

Answer: B

123. Which statement is true about backtracking?

- A) It always uses greedy choice
- B) It explores all possibilities with pruning
- C) It is same as BFS
- D) It never uses recursion

Answer: B

124. In DP, overlapping subproblems means:

- A) Subproblems are independent
- B) Same subproblems are solved repeatedly
- C) Problems are sorted
- D) Problems are unrelated

Answer: B

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125. Which is NOT a property of dynamic programming?

- A) Optimal substructure
- B) Overlapping subproblems
- C) Random selection
- D) Memoization

Answer: C

126. Which sorting algorithm performs best when data is nearly sorted?

- A) Quick sort
- B) Insertion sort
- C) Merge sort
- D) Heap sort

Answer: B

127. Which algorithm guarantees $O(n \log n)$ worst-case time?

- A) Quick sort
- B) Bubble sort
- C) Merge sort
- D) Selection sort

Answer: C

128. Which is true about heap sort?

- A) Uses extra memory $O(n)$
- B) Uses in-place sorting
- C) Uses recursion stack heavily
- D) Always stable

Answer: B

129. Which traversal is used to copy a tree structure?

- A) Inorder
- B) Preorder
- C) Postorder
- D) BFS

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Answer: B

130. Which traversal is used for deleting a tree safely?

- A) Preorder
- B) Inorder
- C) Postorder
- D) Level order

Answer: C

131. In BST, searching worst case occurs when:

- A) Tree is balanced
- B) Tree is skewed
- C) Tree is complete
- D) Tree is AVL

Answer: B

132. AVL tree is a:

- A) Unbalanced BST
- B) Self-balancing BST
- C) Graph structure
- D) Heap structure

Answer: B

133. Rotation in AVL tree is used to maintain:

- A) Sorting order
- B) Height balance
- C) Traversal order
- D) Hashing

Answer: B

134. Which graph representation is better for dense graphs?

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- A) Adjacency list
- B) Adjacency matrix
- C) Stack
- D) Queue

Answer: B

135. Which graph representation is memory efficient for sparse graphs?

- A) Matrix
- B) List
- C) Array of arrays only
- D) Heap

Answer: B

136. DFS uses which principle?

- A) FIFO
- B) LIFO
- C) Random access
- D) Sorting

Answer: B

137. BFS finds shortest path in:

- A) Weighted graph
- B) Unweighted graph
- C) Directed weighted graph
- D) Tree only

Answer: B

138. Dijkstra fails when:

- A) No edges exist
- B) Negative weight edges exist
- C) Graph is undirected
- D) Graph is tree

Answer: B

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139. Bellman-Ford algorithm is slower than Dijkstra because:

- A) Uses recursion
- B) Relaxes edges multiple times
- C) Uses heap
- D) Uses BFS

Answer: B

140. Floyd-Warshall is based on:

- A) Greedy
- B) DP
- C) Backtracking
- D) BFS

Answer: B

141. Time complexity of Floyd-Warshall:

- A) $O(V^2)$
- B) $O(V^3)$
- C) $O(E \log V)$
- D) $O(V + E)$

Answer: B

142. Kruskal algorithm primarily uses:

- A) Sorting edges
- B) BFS
- C) DFS
- D) Stack

Answer: A

143. Prim's algorithm uses:

- A) Queue
- B) Priority queue

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- C) Stack
- D) Array only

Answer: B

144. Which algorithm is greedy?

- A) Merge sort
- B) Dijkstra
- C) DFS
- D) BFS

Answer: B

145. Which algorithm uses divide and conquer?

- A) BFS
- B) DFS
- C) Merge sort
- D) Dijkstra

Answer: C

146. Quick sort pivot worst choice leads to:

- A) $O(n \log n)$
- B) $O(n^2)$
- C) $O(\log n)$
- D) $O(n)$

Answer: B

147. Merge sort requires extra space of:

- A) $O(1)$
- B) $O(n)$
- C) $O(\log n)$
- D) $O(n^2)$

Answer: B

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148. Binary search works on:

- A) Linked list
- B) Sorted array
- C) Graph
- D) Stack

Answer: B

149. Time complexity of linear search:

- A) $O(\log n)$
- B) $O(n)$
- C) $O(n \log n)$
- D) $O(1)$

Answer: B

150. Which is NOT a sorting algorithm?

- A) Quick sort
- B) Merge sort
- C) BFS
- D) Heap sort

Answer: C

151. Which structure supports LRU cache efficiently?

- A) Stack
- B) Queue + HashMap
- C) Tree only
- D) Array only

Answer: B

152. Which data structure is best for priority scheduling?

- A) Stack
- B) Queue
- C) Heap
- D) Linked list

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Answer: C

153. Which is true about recursion?

- A) Always faster than iteration
- B) Uses stack memory
- C) Does not use memory
- D) Avoids function calls

Answer: B

154. Tail recursion means:

- A) Recursion at start
- B) Recursive call is last operation
- C) No base case
- D) Multiple recursive calls

Answer: B

155. Which is better for large datasets?

- A) Bubble sort
- B) Merge sort
- C) Selection sort
- D) Linear search

Answer: B

156. Which algorithm is not stable?

- A) Merge sort
- B) Quick sort
- C) Bubble sort
- D) Insertion sort

Answer: B

157. Which traversal gives minimum height path?

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- A) BFS
- B) DFS
- C) Inorder
- D) Postorder

Answer: A

158. Which structure is used in topological sort?

- A) Stack
- B) Queue
- C) Heap
- D) Array

Answer: A

159. Topological sort is valid only for:

- A) Cyclic graph
- B) DAG
- C) Tree only
- D) Undirected graph

Answer: B

160. Which algorithm is best for cycle detection in undirected graph?

- A) BFS only
- B) DFS with parent tracking
- C) Dijkstra
- D) Prim

Answer: B

161. Which is true about greedy algorithm?

- A) Always optimal globally
- B) Locally optimal choice
- C) Uses DP table
- D) Uses recursion only

Answer: B

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162. Which is a DP problem?

- A) Sorting
- B) Fibonacci
- C) BFS
- D) DFS

Answer: B

163. Which technique reduces recomputation?

- A) Sorting
- B) Memoization
- C) BFS
- D) DFS

Answer: B

164. Which is worst-case complexity of selection sort?

- A) $O(n \log n)$
- B) $O(n^2)$
- C) $O(n)$
- D) $O(\log n)$

Answer: B

165. Which algorithm is used in OS scheduling?

- A) BFS
- B) Heap
- C) Priority queue
- D) DFS

Answer: C

166. Which graph algorithm uses relaxation technique?

- A) DFS
- B) Dijkstra

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- C) Merge sort
- D) BFS

Answer: B

167. Which is true about stack?

- A) FIFO
- B) LIFO
- C) Random access
- D) Sorted structure

Answer: B

168. Which is true about queue?

- A) LIFO
- B) FIFO
- C) Random access
- D) Tree structure

Answer: B

169. Which structure is used in recursion tracking?

- A) Queue
- B) Call stack
- C) Heap
- D) Graph

Answer: B

170. Which algorithm is best for dense graphs MST?

- A) Kruskal
- B) Prim
- C) DFS
- D) BFS

Answer: B

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171. Which algorithm is best for sparse graphs MST?

- A) Prim
- B) Kruskal
- C) DFS
- D) BFS

Answer: B

172. Which sorting algorithm is stable and $O(n \log n)$?

- A) Heap sort
- B) Merge sort
- C) Quick sort
- D) Selection sort

Answer: B

173. Which search is faster in hash table?

- A) $O(n)$
- B) $O(1)$ average
- C) $O(\log n)$
- D) $O(n \log n)$

Answer: B

174. Collision resolution method is:

- A) Sorting
- B) Probing
- C) DFS
- D) BFS

Answer: B

175. Which probing checks next slot sequentially?

- A) Quadratic
- B) Linear
- C) Double hashing
- D) Random

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Answer: B

176. Which is NOT a graph property?

- A) Vertex
- B) Edge
- C) Pivot
- D) Weight

Answer: C

177. Which algorithm uses priority queue?

- A) BFS
- B) Dijkstra
- C) DFS
- D) Merge sort

Answer: B

178. Which is true about heap?

- A) Always sorted
- B) Complete binary tree
- C) Linked list
- D) Graph structure

Answer: B

179. Which algorithm uses recursion heavily?

- A) BFS
- B) DFS
- C) Queue
- D) Hashing

Answer: B

180. Which technique is used in optimal substructure?

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- A) DP
- B) BFS
- C) Sorting
- D) Hashing

Answer: A

181. Which of the following best describes a “greedy choice property”?

- A) Choices depend on future decisions
- B) Locally optimal choice leads to global optimum
- C) All possibilities are explored
- D) Solution is always recursive

Answer: B

182. Which algorithm is most sensitive to pivot selection?

- A) Merge sort
- B) Quick sort
- C) Heap sort
- D) BFS

Answer: B

183. What is the height of a complete binary tree with n nodes?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(n \log n)$
- D) $O(1)$

Answer: B

184. In a BST, which operation is most costly in worst case?

- A) Insert
- B) Search
- C) Traversal
- D) Delete leaf node

Answer: B

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185. Which traversal is used for evaluating expression trees?

- A) Inorder
- B) Preorder
- C) Postorder
- D) Level order

Answer: C

186. Which data structure is used in iterative DFS?

- A) Queue
- B) Stack
- C) Heap
- D) Array only

Answer: B

187. Which algorithm uses edge relaxation repeatedly?

- A) BFS
- B) Dijkstra
- C) Quick sort
- D) DFS

Answer: B

188. What is the main drawback of naive recursion?

- A) Too much sorting
- B) Repeated computation
- C) No loops used
- D) Too much memory allocation always

Answer: B

189. Which technique improves recursion performance?

- A) Backtracking
- B) Memoization

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- C) Heap sort
- D) BFS

Answer: B

190. Which sorting algorithm is in-place but not stable?

- A) Merge sort
- B) Quick sort
- C) Insertion sort
- D) Bubble sort

Answer: B

191. What is the average time complexity of Quick sort?

- A) $O(n^2)$
- B) $O(n \log n)$
- C) $O(n)$
- D) $O(\log n)$

Answer: B

192. Which condition makes Merge sort inefficient compared to Quick sort?

- A) Large dataset
- B) Small dataset due to overhead
- C) Sorted input
- D) Random input

Answer: B

193. Which algorithm is best for finding shortest path in unweighted graphs?

- A) DFS
- B) BFS
- C) Dijkstra
- D) Prim

Answer: B

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194. Which is NOT part of DP design?

- A) Optimal substructure
- B) Overlapping subproblems
- C) Greedy selection
- D) Memoization

Answer: C

195. Which graph type allows cycles?

- A) DAG
- B) Tree
- C) General graph
- D) Binary tree

Answer: C

196. Which structure is best for implementing priority scheduling?

- A) Stack
- B) Queue
- C) Heap
- D) Linked list

Answer: C

197. Which algorithm is used for cycle detection in directed graph?

- A) BFS only
- B) DFS recursion stack
- C) Dijkstra
- D) Prim

Answer: B

198. What is the time complexity of inserting into a heap?

- A) $O(1)$
- B) $O(\log n)$
- C) $O(n)$
- D) $O(n \log n)$

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Answer: B

199. Which structure is used in Kruskal's algorithm?

- A) Queue
- B) Stack
- C) Disjoint Set Union
- D) Array only

Answer: C

200. Which algorithm uses "divide and conquer" repeatedly?

- A) BFS
- B) Merge sort
- C) Dijkstra
- D) DFS

Answer: B

201. What is the key idea behind backtracking?

- A) Choose one optimal path only
- B) Try all possibilities and undo invalid ones
- C) Use sorting first
- D) Use hashing only

Answer: B

202. Which algorithm has exponential time complexity in worst case?

- A) Binary search
- B) Fibonacci recursion
- C) Merge sort
- D) BFS

Answer: B

203. Which is true about adjacency matrix?

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- A) Efficient for sparse graphs
- B) Uses $O(V^2)$ space
- C) Uses $O(V+E)$ space
- D) Cannot represent directed graphs

Answer: B

204. Which traversal is iterative by default?

- A) DFS
- B) BFS
- C) Inorder
- D) Preorder recursive only

Answer: B

205. Which algorithm guarantees shortest path in weighted graph with non-negative edges?

- A) BFS
- B) Dijkstra
- C) DFS
- D) Kruskal

Answer: B

206. What is the main limitation of BFS?

- A) Cannot find shortest path
- B) High memory usage
- C) Cannot traverse graphs
- D) Uses recursion always

Answer: B

207. Which sorting algorithm is best for linked lists?

- A) Quick sort
- B) Merge sort
- C) Heap sort
- D) Selection sort

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Answer: B

208. Which is true for recursion stack?

- A) Always constant size
- B) Grows with recursion depth
- C) Independent of function calls
- D) Uses heap memory only

Answer: B

209. Which algorithm is NOT comparison-based?

- A) Merge sort
- B) Quick sort
- C) Radix sort
- D) Heap sort

Answer: C

210. Which is the best case complexity of Quick sort?

- A) $O(n^2)$
- B) $O(n \log n)$
- C) $O(n)$
- D) $O(\log n)$

Answer: B

211. Which technique is used in greedy algorithms?

- A) Local optimization
- B) Backtracking
- C) Memoization
- D) Sorting only

Answer: A

212. Which structure is used in BFS traversal?

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- A) Stack
- B) Queue
- C) Heap
- D) Tree

Answer: B

213. Which graph algorithm is used to detect minimum spanning tree?

- A) DFS
- B) Kruskal
- C) BFS
- D) Binary search

Answer: B

214. Which statement is true about dynamic programming?

- A) Never uses recursion
- B) Stores intermediate results
- C) Always greedy
- D) Uses sorting only

Answer: B

215. Which is worst-case complexity of BFS?

- A) $O(V + E)$
- B) $O(V^2)$
- C) $O(E \log V)$
- D) $O(\log V)$

Answer: A

216. Which is used in Dijkstra algorithm?

- A) Stack
- B) Priority queue
- C) Array only
- D) Linked list only

Answer: B

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217. Which structure is used for recursion calls?

- A) Queue
- B) Stack
- C) Heap
- D) Graph

Answer: B

218. Which algorithm is best for solving knapsack problem (optimal)?

- A) Greedy
- B) Dynamic programming
- C) BFS
- D) DFS

Answer: B

219. Which is true about greedy algorithms?

- A) Always give optimal solution
- B) May fail for global optimum
- C) Always recursive
- D) Always DP-based

Answer: B

220. Which graph traversal is best for level-order processing?

- A) DFS
- B) BFS
- C) Inorder
- D) Postorder

Answer: B

221. Which condition guarantees that a recursive function will terminate?

- A) Multiple recursive calls
- B) Presence of base case

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- C) Use of arrays
- D) Use of loops

Answer: B

222. What is the main advantage of using a stack over recursion?

- A) Faster sorting
- B) No memory usage
- C) Avoids function call overhead
- D) Always uses less space

Answer: C

223. Which case causes binary search to fail?

- A) Sorted array
- B) Duplicate elements
- C) Unsorted array
- D) Even number of elements

Answer: C

224. What is the height of a skewed binary tree with n nodes?

- A) $O(\log n)$
- B) $O(n)$
- C) $O(1)$
- D) $O(n \log n)$

Answer: B

225. Which data structure is best for implementing recursion explicitly?

- A) Queue
- B) Stack
- C) Heap
- D) Graph

Answer: B

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226. Which traversal uses FIFO logic?

- A) DFS
- B) BFS
- C) Inorder
- D) Postorder

Answer: B

227. Which sorting algorithm is stable and in-place?

- A) Merge sort
- B) Quick sort
- C) Bubble sort
- D) Heap sort

Answer: C

228. What is worst-case time complexity of Bubble Sort?

- A) $O(n)$
- B) $O(n \log n)$
- C) $O(n^2)$
- D) $O(\log n)$

Answer: C

229. Which algorithm uses divide and conquer strategy?

- A) BFS
- B) Quick sort
- C) DFS
- D) Linear search

Answer: B

230. What is space complexity of BFS?

- A) $O(1)$
- B) $O(V)$
- C) $O(\log V)$
- D) $O(E)$ only

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Answer: B

231. Which is NOT a property of a tree?

- A) Acyclic
- B) Connected
- C) Cyclic
- D) Hierarchical

Answer: C

232. Which graph traversal is recursive by nature?

- A) BFS
- B) DFS
- C) Level order
- D) None

Answer: B

233. Which algorithm is best for dense graphs MST?

- A) Kruskal
- B) Prim
- C) BFS
- D) DFS

Answer: B

234. Which structure is used in Kruskal's algorithm?

- A) Stack
- B) Disjoint set
- C) Queue
- D) Array only

Answer: B

235. What is the time complexity of DFS?

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- A) $O(V + E)$
- B) $O(V^2)$
- C) $O(E^2)$
- D) $O(\log V)$

Answer: A

236. Which algorithm uses greedy choice at each step?

- A) Merge sort
- B) Dijkstra
- C) BFS
- D) DFS

Answer: B

237. Which sorting algorithm is best for small arrays?

- A) Quick sort
- B) Insertion sort
- C) Merge sort
- D) Heap sort

Answer: B

238. What is the worst-case of Quick Sort?

- A) $O(n \log n)$
- B) $O(n^2)$
- C) $O(n)$
- D) $O(\log n)$

Answer: B

239. Which structure is best for LRU cache?

- A) Stack
- B) Queue + HashMap
- C) Tree
- D) Array

Answer: B

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240. Which algorithm is used for shortest path with negative weights?

- A) Dijkstra
- B) BFS
- C) Bellman-Ford
- D) Prim

Answer: C

241. Which traversal is used to evaluate postfix expression tree?

- A) Inorder
- B) Preorder
- C) Postorder
- D) Level order

Answer: C

242. Which is true about recursion?

- A) Always faster than iteration
- B) Uses call stack
- C) Does not use memory
- D) Avoids function calls

Answer: B

243. Which algorithm is NOT comparison-based?

- A) Quick sort
- B) Merge sort
- C) Radix sort
- D) Heap sort

Answer: C

244. Which structure is used in BFS?

- A) Stack
- B) Queue

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- C) Heap
- D) Tree

Answer: B

245. Which algorithm is best for sparse graph MST?

- A) Prim
- B) Kruskal
- C) DFS
- D) BFS

Answer: B

246. Which technique avoids recomputation?

- A) Greedy
- B) Memoization
- C) BFS
- D) Sorting

Answer: B

247. What is optimal substructure?

- A) Problem solved randomly
- B) Optimal solution from subproblems
- C) Sorting property
- D) Graph property

Answer: B

248. Which algorithm is used in binary search?

- A) Greedy
- B) Divide and conquer
- C) DP
- D) Backtracking

Answer: B

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249. Which structure is best for DFS?

- A) Queue
- B) Stack
- C) Heap
- D) Array

Answer: B

250. Which sorting algorithm is stable?

- A) Quick sort
- B) Heap sort
- C) Merge sort
- D) Selection sort

Answer: C

251. What is time complexity of binary search?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(n \log n)$
- D) $O(1)$

Answer: B

252. Which is worst case for binary search?

- A) Empty array
- B) Sorted array
- C) Element not present
- D) Reverse array

Answer: C

253. Which structure is used in recursion?

- A) Queue
- B) Stack
- C) Tree
- D) Graph

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Answer: B

254. Which graph algorithm uses priority queue?

- A) BFS
- B) DFS
- C) Dijkstra
- D) Inorder

Answer: C

255. Which is true for heap?

- A) Always sorted
- B) Complete binary tree
- C) Linear structure
- D) Graph structure

Answer: B

256. Which traversal visits level by level?

- A) DFS
- B) BFS
- C) Inorder
- D) Postorder

Answer: B

257. Which algorithm is used for cycle detection?

- A) BFS only
- B) DFS
- C) Dijkstra
- D) Merge sort

Answer: B

258. Which algorithm is used for topological sorting?

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- A) BFS only
- B) DFS
- C) Heap sort
- D) Quick sort

Answer: B

259. Which algorithm is greedy?

- A) Merge sort
- B) Dijkstra
- C) DFS
- D) BFS

Answer: B

260. Which technique is used in DP?

- A) Random choice
- B) Overlapping subproblems
- C) Sorting only
- D) Graph only

Answer: B

261. Which is NOT a graph traversal?

- A) BFS
- B) DFS
- C) Inorder
- D) Level order

Answer: C

262. Which algorithm is best for shortest path in unweighted graph?

- A) DFS
- B) BFS
- C) Dijkstra
- D) Prim

Answer: B

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263. Which is true about Quick sort?

- A) Stable
- B) In-place
- C) Always $O(n \log n)$ worst
- D) Uses queue

Answer: B

264. Which algorithm uses recursion heavily?

- A) BFS
- B) DFS
- C) Queue
- D) Hashing

Answer: B

265. Which is best case of merge sort?

- A) $O(n^2)$
- B) $O(n \log n)$
- C) $O(n)$
- D) $O(\log n)$

Answer: B

266. Which structure is used in DFS recursion?

- A) Queue
- B) Stack
- C) Heap
- D) Array

Answer: B

267. Which algorithm is used in MST?

- A) BFS
- B) Prim

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- C) DFS
- D) Binary search

Answer: B

268. Which algorithm is used in MST (alternative)?

- A) Kruskal
- B) BFS
- C) DFS
- D) Sorting only

Answer: A

269. Which is true about adjacency matrix?

- A) Sparse efficient
- B) Dense efficient
- C) Uses $O(V+E)$
- D) Uses stack

Answer: B

270. Which is true about adjacency list?

- A) $O(V^2)$ space
- B) $O(V+E)$ space
- C) $O(n^2 \log n)$
- D) $O(1)$ space

Answer: B

271. Which algorithm is best for large sparse graph MST?

- A) Prim
- B) Kruskal
- C) DFS
- D) BFS

Answer: B

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272. Which is true about DP?

- A) Always recursive
- B) Uses table storage
- C) No optimization
- D) Only greedy

Answer: B

273. Which is NOT a DP problem?

- A) Fibonacci
- B) Knapsack
- C) BFS
- D) LCS

Answer: C

274. Which is used in Dijkstra?

- A) Stack
- B) Priority queue
- C) Queue
- D) Array only

Answer: B

275. Which is true about BFS memory?

- A) Low memory
- B) High memory
- C) No memory
- D) Constant memory

Answer: B

276. Which sorting is best for nearly sorted data?

- A) Merge sort
- B) Insertion sort
- C) Heap sort
- D) Quick sort worst

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Answer: B

277. Which algorithm is brute force?

- A) Quick sort
- B) Bubble sort
- C) Merge sort
- D) Dijkstra

Answer: B

278. Which structure is used in recursion?

- A) Queue
- B) Call stack
- C) Heap
- D) Graph

Answer: B

279. Which algorithm is used for shortest path all pairs?

- A) Dijkstra
- B) BFS
- C) Floyd-Warshall
- D) Prim

Answer: C

280. Which is true about greedy?

- A) Always optimal
- B) Locally optimal choice
- C) Uses DP table
- D) Uses recursion only

Answer: B

281. Which algorithm is divide and conquer?

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- A) BFS
- B) Merge sort
- C) DFS
- D) Dijkstra

Answer: B

282. Which is true about stack?

- A) FIFO
- B) LIFO
- C) Random
- D) Sorted

Answer: B

283. Which is true about queue?

- A) LIFO
- B) FIFO
- C) Random
- D) Tree

Answer: B

284. Which algorithm is used in binary tree traversal?

- A) BFS only
- B) DFS only
- C) Both BFS and DFS
- D) Sorting

Answer: C

285. Which is true about recursion depth?

- A) Constant
- B) Depends on input
- C) Always 1
- D) Always infinite

Answer: B

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286. Which algorithm is used for scheduling?

- A) BFS
- B) Priority queue
- C) DFS
- D) Merge sort

Answer: B

287. Which is true about heap?

- A) Always sorted
- B) Parent-child relation
- C) Linear list
- D) Graph

Answer: B

288. Which algorithm is used for cycle detection in undirected graph?

- A) BFS only
- B) DFS with parent tracking
- C) Dijkstra
- D) Prim

Answer: B

289. Which is true about DP?

- A) No recursion ever
- B) Stores results
- C) Always greedy
- D) Always sorting

Answer: B

290. Which is true about backtracking?

- A) Ignores constraints
- B) Explores all possibilities

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- C) Always greedy
- D) Only sorting

Answer: B

291. Which algorithm is used in path finding?

- A) BFS
- B) DFS
- C) Dijkstra
- D) All of these depending on case

Answer: D

292. Which is true about Quick sort?

- A) Always stable
- B) Uses pivot
- C) Uses queue
- D) O(n) always

Answer: B

293. Which is best for large dataset sorting?

- A) Bubble sort
- B) Merge sort
- C) Selection sort
- D) Linear search

Answer: B

294. Which structure is used in graph BFS?

- A) Stack
- B) Queue
- C) Heap
- D) Tree

Answer: B

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295. Which algorithm is used in shortest path DAG?

- A) BFS
- B) Topological sort + DP
- C) DFS only
- D) Heap only

Answer: B

296. Which is true about DFS?

- A) Level order
- B) Deep traversal
- C) FIFO
- D) Sorting

Answer: B

297. Which is true about BFS?

- A) Depth first
- B) Level order
- C) Recursive only
- D) Sorting

Answer: B

298. Which is best for recursion replacement?

- A) Queue
- B) Stack
- C) Graph
- D) Array

Answer: B

299. Which algorithm is used for MST?

- A) BFS
- B) Prim
- C) DFS
- D) Binary search

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Answer: B

300. Which technique is most general for optimization problems?

- A) Greedy only
- B) Dynamic programming
- C) BFS
- D) DFS

Answer: B
